Using (4) in (3) leads to $\tan^{-1}\left(\frac{y-x}{1+xy}\right) = C$, which implies $\frac{y-x}{1+xy} = \tan C = a$ (say).

Thus, $y-x = a(1+xy)$ gives the required solution.

Example 10.12

Find the particular solution of $\left(1+x^3\right)dy - x^2 y dx = 0$ satisfying the condition $y(1) = 2$.

Solution

Given that $(1+x^3)dy - x^2 y dx = 0$.

The above equation is written as $\frac{dy}{y} - \frac{x^2}{1+x^3}dx = 0$.

Integrating both sides gives $\log y - \frac{1}{3}\log(1+x^3) = C_1$, which implies,

$$3\log y - \log(1+x^3) = \log C.$$

$$\text{Thus, } 3\log y = \log(1+x^3) + \log C,$$

$$\text{which reduces to } \log y^3 = \log C(1+x^3).$$

Hence, $y^3 = C(1+x^3)$ gives the general solution of the given differential equation. It is given that when $x=1,\ y=2$. Then $2^3 = C(1+1) \implies C = 4$ and hence the particular solution is $y^3 = 4(1+x^3)$.

10.6.2 Substitution Method

Let the differential equation be of the form $\frac{dy}{dx} = f(ax+by+c)$.

(i) If $a \neq 0$ and $b \neq 0$, then the substitution $ax+by+c = z$ reduces the given equation to the variables separable form.

(ii) If $a = 0$ or $b = 0$, then the differential equation is already in separable form.

Example 10.13

Solve $y' = \sin^2\left(x-y+1\right)$.

Solution

Given that $y' = \sin^2\left(x-y+1\right)$

$$\text{Put } z = x-y+1, \text{ so that } \frac{dz}{dx} = 1 - \frac{dy}{dx}.$$

Thus, the given equation reduces to $1 - \frac{dz}{dx} = \sin^2 z$.

$$\text{i.e., } \frac{dz}{dx} = 1 - \sin^2 z = \cos^2 z.$$

$$\text{Separating the variables leads to } \frac{dz}{\cos^2 z} = dx \text{ (or) } \sec^2 z dz = dx.$$

$$\text{On integration, we get } \tan z = x + C \text{ (or) } \tan\left(x-y+1\right) = x + C.$$

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